

Measuring and Discussing Computer System Performance

or

“My computer is faster than your computer”

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Dean Tullsen

The bottom line: Performance

Car	Time to Bay Area	Speed	Passengers	Throughput (pmp)
Ferrari	3.1 hours	160 mph	2	320
Greyhound	7.7 hours	65 mph	60	3900

° Time to do the task

– , response time, latency

° Tasks per day, hour, week, sec, ns. ..

– , bandwidth

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How to measure Execution Time?

```
% time program
... program results ...
90.7u 12.9s 2:39 65%
%
```

- Wall-clock time?
- user CPU time?
- user + kernel CPU time?
- Answer:

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Our definition of Performance

$$\text{Performance}_X = \frac{1}{\text{Execution Time}_X}, \text{ for program } X$$

- only has meaning in the context of a or
- Not very intuitive as an absolute measure, but most of the time we're more interested in **relative performance**.

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Relative Performance, the Definition

$$\text{Relative Performance}_{(X/Y)} = \frac{\text{Performance}_X}{\text{Performance}_Y} = \frac{\text{Execution Time}_Y}{\text{Execution Time}_X} = n$$

Example

- Machine A runs program C in 9 seconds, Machine B runs the same program in 6 seconds. What is the speedup we see if we move to Machine B from Machine A?
- Machine B gets a new compiler, and can now run the program in 3 seconds. ???

What is Time?

CPU Execution Time = CPU clock cycles * Clock cycle time

- Every conventional processor has a clock with an associated clock cycle time or clock rate
- Every program runs in an integral number of clock cycles

Cycle Time

MHz = millions of cycles/second, GHz = billions of cycles/second

X MHz = 1000/X nanoseconds cycle time

Y GHz = 1/Y nanoseconds cycle time

How many clock cycles?

Number of CPU cycles = Instructions executed * Average Clock Cycles per Instruction (CPI)

All Together Now

$$\text{CPU Execution Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

Diagram illustrating the units for each term in the equation:

- Instruction Count**: instructions
- CPI**: cycles/instruction
- Clock Cycle Time**: seconds/cycle
- CPU Execution Time**: seconds

$$\text{CPU Execution Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

- IC = 1 billion, 500 MHz processor, execution time of 3 seconds. What is the CPI for this program?
- Suppose we reduce CPI to 1.2 (through an architectural improvement). What is the new execution time?

Who Affects Performance?

$$\text{CPU Execution Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

- programmer
- compiler
- instruction-set architect
- machine architect
- hardware designer
- materials scientist/physicist/silicon engineer

Performance Variation

$$\text{CPU Execution Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

	Number of instructions	CPI	Clock Cycle Time
Same machine different programs			
same programs, different machines, same ISA			
Same programs, different machines			

Other Performance Metrics

- MIPS
- MFLOPS

MIPS

MIPS = Millions of Instructions Per Second

$$= \frac{\text{Instruction Count}}{\text{Execution Time} * 10^6}$$

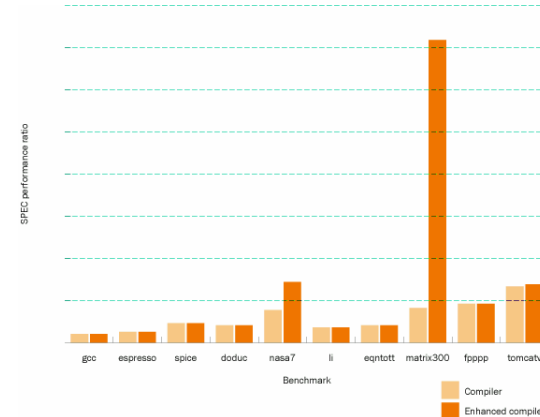
$$= \frac{\text{Clock rate}}{\text{CPI} * 10^6}$$

- program-independent
- deceptive

Which Programs?

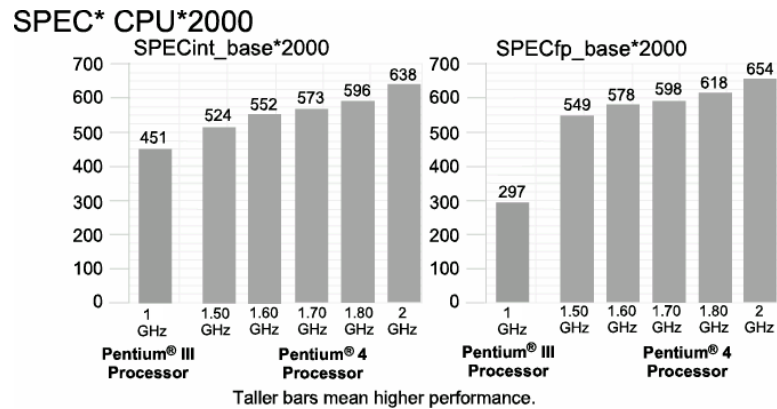
- peak throughput measures (simple programs)?
- synthetic benchmarks (whetstone, dhrystone,...)?
- Real applications
- SPEC (best of both worlds, but with problems of their own)
 - *System Performance Evaluation Cooperative*
 - Provides a common set of real applications along with strict guidelines for how to run them.
 - provides a relatively unbiased means to compare machines.

Danger in Benchmark-Specific Performance Measures



- measures as much as **architecture!**

SPEC Performance on Pentium III and Pentium 4



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Amdahl's Law

- The impact of a performance improvement is limited by the percent of execution time affected by the improvement

$$\text{Execution time after improvement} = \frac{\text{Execution Time Affected}}{\text{Amount of Improvement}} + \text{Execution Time Unaffected}$$

- Make the fast!!

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Key Points

- Be careful how you specify performance
- Execution time = instructions * CPI * cycle time
- Use real applications
- Use standards, if possible
- Make the common case fast

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